MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY, HIGH VOLTAGE
TYPES 1N6528 THROUGH 1N6535
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, high voltage, fast recovery semiconductor diodes for use as power rectifiers. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Types	V_{RWM}	I_O T_A = 55°C 1/	I_O T_A = 100°C 2/	I_{FSM} t_p = 8.3 ms	t_{rr}	T_{STG}	T_J	$R_{\theta JL}$ L = .25
	V dc	mA dc	mA dc	A (pk)	ns	°C -65 to +200	°C -65 to +175	°C/W
1N6528	1500	250	125	10	70			50
1N6529	2000	250	125	10	70			50
1N6530	2500	100	50	8	70			50
1N6531	3000	100	50	8	70			50
1N6532	4000	50	25	4	70			50
1N6533	5000	50	25	4	70			50
1N6534	7500	25	12.5	2	70			50
1N6535	10,000	25	12.5	2	70			50

1/ Derate linearly for 55°C $\leq T_A \leq$ 100°C. I_O @ T_A = +55°C to I_O @ T_A = +100°C.

2/ Derate linearly for 100°C $\leq T_A \leq$ 175°C. I_O @ T_A = +100°C to I_O = 0 A @ T_A = +175°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter

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1.4 Primary electrical characteristics.

Types	V_{RWM}	I_O T_A = 55°C	I_{R1} T_A = 25°C	V_{F1}	C V_R = 50 V F_O = 1kHz
	V dc	mA dc	µA dc	V (pk)	pF
1N6528	1500	250	0.10	3.0	4.0
1N6529	2000	250	0.10	3.0	4.0
1N6530	2500	100	0.10	7.0	2.0
1N6531	3000	100	0.10	7.0	2.0
1N6532	4000	50	0.10	9.0	1.0
1N6533	5000	50	0.10	9.0	1.0
1N6534	7500	25	0.10	14.0	0.5
1N6535	10000	25	0.10	14.0	0.5

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-1276 - Leads for Electronic Component Parts.

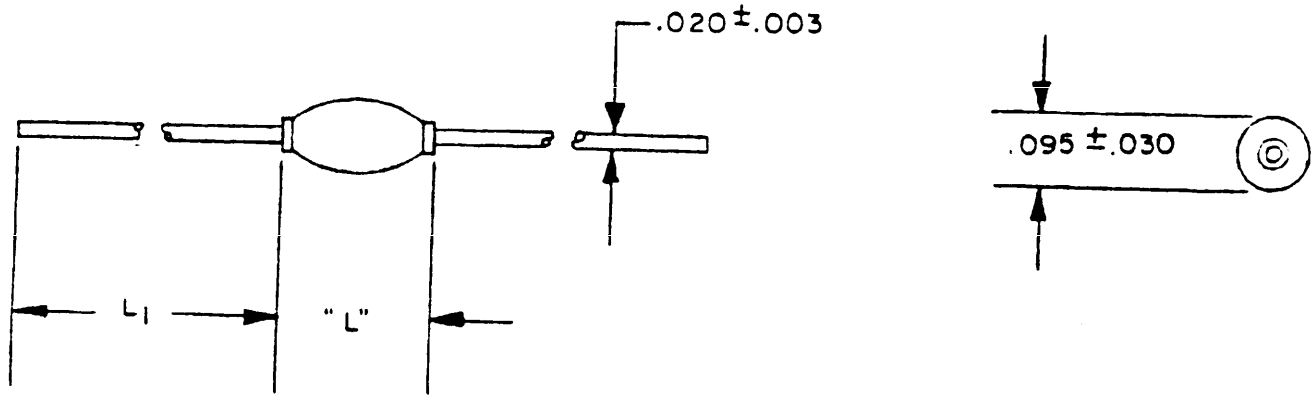
(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.



Part no.	L inches		L_1 inches	
	min	max	min	max
1N6528	.140	.200	1.0	1.3
1N6529	.140	.200	"	"
1N6530	.160	.220	"	"
1N6531	.160	.220	"	"
1N6532	.180	.240	"	"
1N6533	.180	.240	"	"
1N6534	.240	.300	"	"
1N6535	.240	.300	"	"

Inches	mm
.003	0.08
.020	0.50
.030	0.76
.095	2.41
.140	3.56
.180	4.57
.200	5.08
.220	5.58
.240	6.10
.280	7.11
1.00	25.4
1.3	33.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 1. Dimensions and configuration.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1.

3.3.1 Lead material and finish. Lead material shall be type C, 99.9 percent silver or copper in accordance with MIL-STD-1276. Lead finish shall be gold, tin or solder dipped. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the following marking may be omitted from the body of the device:

a. Manufacturer's identification.
b. Country of origin.
c. Lot identification code.
d. "1N" of the type designation.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTXV and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-STD-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Surge, see 4.3.2	Surge, see 4.3.2
9	V_{F1} and I_{R1}	Not applicable
11	V_{F1} and I_{R1}; ΔV_{F1} and ΔI_{R1}, see table V.	V_{F1} and I_{R1}
12	See 4.3.1	See 4.3.1.
13	Subgroup 2 and 3 of table I herein; ΔV_{F1} and ΔI_{R1} see table V V_{F1}, I_{R1}	Subgroup 2 of table I herein; ΔV_{F1} and ΔI_{R1} see table V V_{F1}, I_{R1}

1/ Surge screening shall be performed anytime after screen 3 and before screen 10.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

Types	V_{RWM} (V dc)	I_O (mA dc)	T_A °C (min)	f (Hz)
1N6528	1000	250	25	≥ 60
1N6529	"	250	"	"
1N6530	"	100	"	"
1N6531	"	100	"	"
1N6532	"	50	"	"
1N6533	"	50	"	"
1N6534	"	25	"	"
1N6535	"	25	"	"

4.3.2 Surge screening. See MIL-STD-750, method 4066. V_{RWM} = 0. Six surges, T_A = 25°C. Apply 20 x I_O at 55°C; t_p = 8.3 ms.

4.4 Quality conformance inspection. Quality conformance inspections shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspections shall be conducted in accordance with MIL-S-19500, and table I herein. Endpoint electrical measurements shall be in accordance with the applicable steps of table V herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

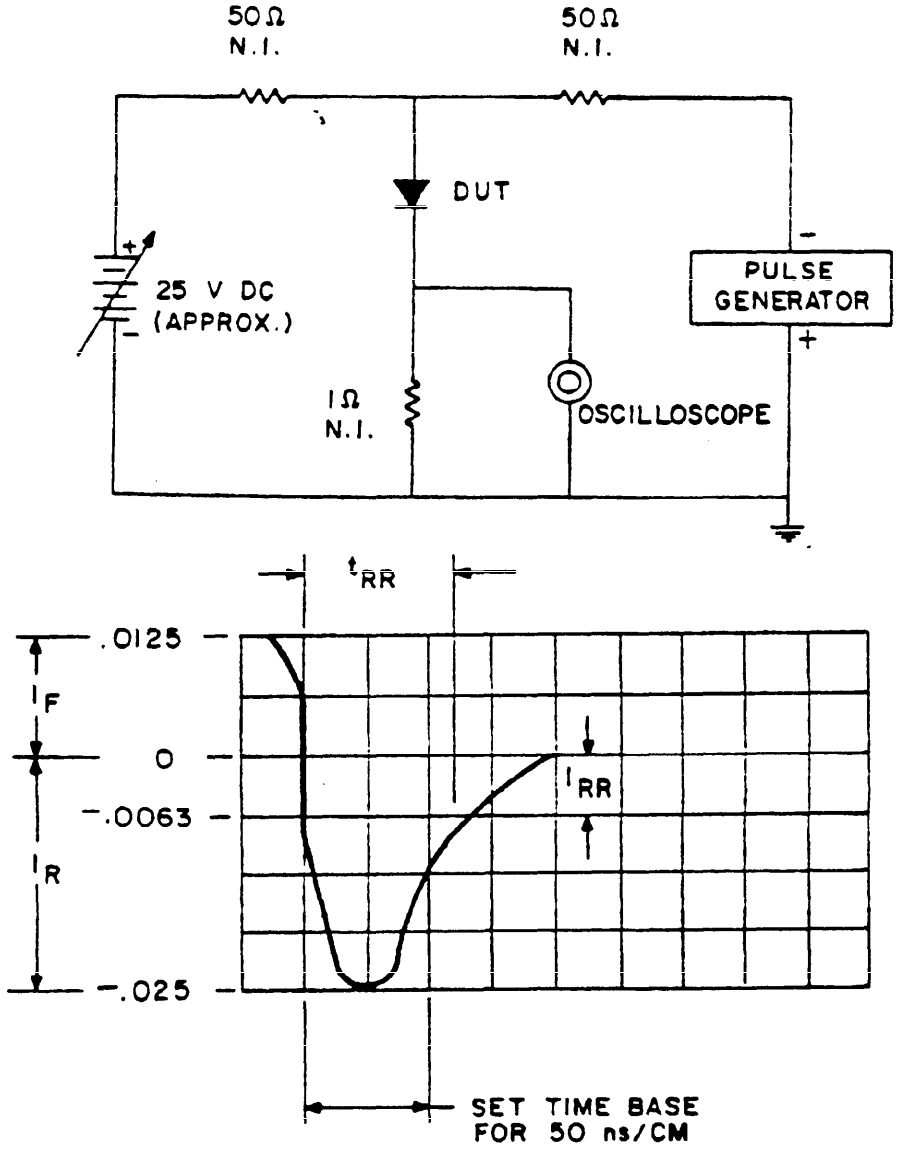
4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.4 Group E inspection. Group E inspections shall be conducted in accordance with table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Reverse recovery time. The reverse-recovery time shall be measured in the circuit of figure 2 or an equivalent circuit. The recovery conditions shall be 12.5 mA forward current to 25 mA reverse current. The reverse-recovery time is defined as the time the rectifier begins to conduct in the reverse direction (crosses I = 0) until the reverse current decays to -6.25 mA. The point of contact on the leads shall be no less than .375 inch from the diode body.



NOTES:
1. Oscilloscope-rise time ≤ 7 ns; input impedance = 1 megohm; 22 pF.
2. Pulse generator - rise time ≤ 10 ns; source impedance 50Ω.
3. Recovery time shall be measured on the above circuit and with equipment as shown. The pulse generator shall have a pulse repetition frequency of 1 HKz and a pulse width of 200 ns recovery conditions 12.5 A forward current to 25 mA reverse current. Recovery time measured when rectifier recovers to 6.3 mA.

FIGURE 2. Reverse-recovery-time test circuit and characteristic nomograph.

TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 3 Continued							
Low temperature operation:		T_A = -55°C					
Forward voltage	4011	I_F = 25 mA		V_{F2}			
1N6528						4.8	V dc
1N6529						4.8	V dc
1N6530						11.2	V dc
1N6531						11.2	V dc
1N6532						14.4	V dc
1N6533						14.4	V dc
1N6534						22.4	V dc
1N6535						22.4	V dc
Breakdown voltage	4021	I_R = 50 µA		$V_{(BR)R2}$			
1N6528					1500		V dc
1N6529					2000		V dc
1N6530					2500		V dc
1N6531					3000		V dc
1N6532					4000		V dc
1N6533					5000		V dc
1N6534					7500		V dc
1N6535					10000		V dc
Subgroup 4							
Reverse recovery time		See figure 2 and 4.5.2.		t_{rr}		70	ns
Capacitance	4001	V_R = 50 V dc 1 kHz ≤ f ≤ 100 kHz		C			
1N6528						4.0	pF
1N6529						4.0	pF
1N6530						2.0	pF
1N6531						2.0	pF
1N6532						1.0	pF
1N6533						1.0	pF
1N6534						0.5	pF
1N6535						0.5	pF
Subgroups 5, 6, and 7							
Not applicable							

1/ LTPD numbers used are taken from MIL-S-19500.

TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1.		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Surge current	4066	I_O = I_O at T_A of 55°C I_{FSM} = see 1.3; V_{RWM} = 0 V one surge 8.3 ms		
Hermetic seal	1071			
Fine leak Gross leak				
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, and 6.		
Decap internal visual	2075			
Subgroup 4				
Intermittent operation life	1037	See 4.3.1 t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles.		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, and 6.		
Subgroup 5				
Accelerated steady-state operation life	1027	T_A = 150°C min., I_O = (see 1.3) min, or adjust T_A or I_O as required by the corresponding chosen value of T_A or I_O to achieve a lot T_J = 275°C for a minimum of 96 hours at V_{RWM} = 1,000 V.		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7.		
Subgroup 6				
Thermal resistance	4081	T_A = 25°C; $R_{\theta JL}$ = see 1.3.		

1/ For LTPD numbers, see MIL-S-19500.

TABLE IIb. Group B inspection for JANTX and JANTXV.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD 1/
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Surge current	4066	I_{FSM} = see 1.3. I_O = I_O at T_A of 55°C V_{RWM} = 0 V, one surge = 8.3 ms	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table V, steps 1, 2, and 3.	
Subgroup 3			
Steady-state dc operation life	1027	See 4.3.1	
Electrical measurements		See table V, steps 1, 2, 3, and 4.	
Subgroup 4			
Not applicable			
Subgroup 5			
Thermal resistance	4081	T_j = 25°C; $R_{\theta JL}$ = (see 1.3). $R_{\theta JT}$ = see 1.3.	
Subgroup 6			
High-temperature life (nonoperating)	1032	T_A = 200°C	
Electrical measurements		See table V, steps 1, 2, 3, and 4.	

1/ For LTPD numbers, see MIL-S-19500.

TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Physical dimensions	2066	See figure 1.	
Subgroup 2			
Thermal shock (glass strain)	1056		
Terminal strength	2036	Test condition A; weight = 5 lbs; t = 30 s	
Hermetic seal	1071		
Fine leak Gross leak			
Moisture resistance	1021		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7 (JANS), and steps 2 and 3 (JANTX and JANTXV).	
Subgroup 3			
Shock	2016		
Vibration, variable frequency	2056	Nonoperating	
Constant acceleration	2006		
Electrical measurements		See table V, steps 1, 2 and 3 (all levels).	
Subgroup 4			
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Barometric pressure (reduced)	1001	Pressure = 8.0 mm Hg, t = 1 minute (min).	
Subgroup 6			
Steady-state dc operation life	1026	See 4.3.1.	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, and 7 (JANS), and steps 2, 3, 4, and 5 (JANTX and JANTXV).	

1/ For LTPD numbers, see MIL-S-19500.

TABLE IV. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		LTPD	Small lot quality conformance inspection n/c
	Method	Conditions		
Subgroup 1			5	12/0
Thermal shock	1051	500 cycles		
Electrical measurement		See table V, steps 1 and 2		
Subgroup 2			5	12/0
Steady-state dc blocking life	1038	1,000 hours, condition A.		
Electrical measurement		See table V steps 1, 2		
Subgroup 3			3/0	1/0
Destructive physical analysis		Photos of cross sections shall be submitted in the qualification report. Vendors shall retain duplicate photos		

TABLE V. Groups A, B, and C end-point electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Forward voltage	4011	I_F = 25 mA (pulsed);	V_{F1}			V (pk)
	1N6528		tp = 300 µs; PPR = 60 Hz			3.0	
	1N6529					3.0	
	1N6530					7.0	
	1N6531					7.0	
	1N6532					9.0	
	1N6533					9.0	
	1N6534					14.0	
	1N6535					14.0	
2.	Reverse current	4016	DC method; V_R = (see 1.3)	I_{R1}		0.10	µA dc
3	Reverse recovery		See 4.5.2	t_{rr}		70	ns
4	Scope display						
5	Capacitance	4001	V_R = 50 V dc 1 kHz ≤ f ≤ 100 kHz	C			pF
	1N6528					4.0	
	1N6529					4.0	
	1N6530					2.0	
	1N6531					2.0	
	1N6532					1.0	
	1N6533					1.0	
	1N6534					0.5	
	1N6535					0.5	
6	Forward voltage	4011	I_F = 25 mA dc	ΔV_{F1} 1/			V dc
	1N6528					±0.2	
	1N6529					±0.2	
	1N6530					±0.4	
	1N6531					±0.4	
	1N6532					±0.8	
	1N6533					±0.8	
	1N6534					±1.2	
	1N6535					±1.2	
7	Reverse current	4016	DC method	ΔI_{R1} 1/		50 nA dc or 100% of initial value whichever is greater	

1/ Devices that exceed group A limits for this test shall be rejected

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead material and finish may be specified (see 3.3.1).

6.2.1 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Numbers (PIN's). This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
1N6528	53711	8502234-215
	60211	RM115
		M15FG
		M15UFG
1N6529	53711	8502234-220
	49956	G339806
	94117	194087
	60211	RM116
		RM135
		RM140
		M20FG
		M20UFG
1N6530	53711	8502234-225
	60211	RM117
		M25FG
		M25UFG
1N6531	53711	4027428-103
		4027428-113
		8502234-230
		4056502
	60211	RA352
		RA643
		RM118
		RM123
		M30FG
		M30UFG

Military PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
1N6532	53711	8502234-240
	23426	28005-12
	60211	RM119
		RM130
		M40FG
		M40UFG
1N6533	53711	4027428-105
		4027428-115
		8502234-250
		304-1-58A2
	60211	RA641
		RA644
		RM120
		RM137
		M50FG
		M50UFG
1N6534	53711	402742B-106
		4027428-116
		8502234-260
	23426	28005-7
	60211	RA642
		RA645
		RM121
		RM131
		M60FG
		M60UFG
1N6535	58260	13084424
	53711	8502234-280
		8502234-300
	23426	28005-8
	60211	M100FG
		RM122
		RM109
	23426	RM132
	60211	M100UFG
		M80FG
		M80UFG

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI
Air Force - 11, 19, 85
DLA - ES
NASA - MSFC-EG02

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13
NASA - NA

Preparing activity
Army - ER

Agent:
DLA - ES

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